

Cricket Player Performance Tracking Application: CRICKSTATS

A project Report submitted of the partial fulfilment of the requirements for the
award of the degree of

BACHELOR OF COMPUTER APPLICATIONS

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**DEPARTMENT OF COMPUTER APPLICATIONS
ISLAMIAH WOMEN'S ARTS AND SCIENCE COLLEGE**

(Accredited by NAAC with 'B' grade)

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Affiliated with Thiruvalluvar University VANIYAMBADI-635 752**

Cricket Player Performance Tracking

Application: CRICKSTATS

User Story

As a project team, we want to develop a comprehensive Cricket Player Performance Tracking Application within Salesforce that enables coaches, team managers, and analysts to efficiently record, manage, and evaluate player performance data. The application will utilize custom objects such as Player and Match Performance, along with well-defined relationships, fields, and automation features to ensure accurate data management and easy accessibility. By implementing Salesforce reports, dashboards, flows, and analytics, the system will provide meaningful insights into player and team performance, support informed decision-making, and contribute to effective player development and team management.

1. INTRODUCTION

1.1 Project Overview

CRICKSTATS is a Salesforce-based team project developed to manage, monitor, and analyse cricket player performance through a centralized digital platform. The application leverages Salesforce custom objects, relationships, automation tools, reports, and dashboards to organize match-by-match performance data in a structured and efficient manner.

The system allows coaches, team managers, analysts, and administrators to maintain player records, track performance statistics, generate analytical reports, and gain valuable insights that support strategic planning and performance improvement.

1.2 Purpose

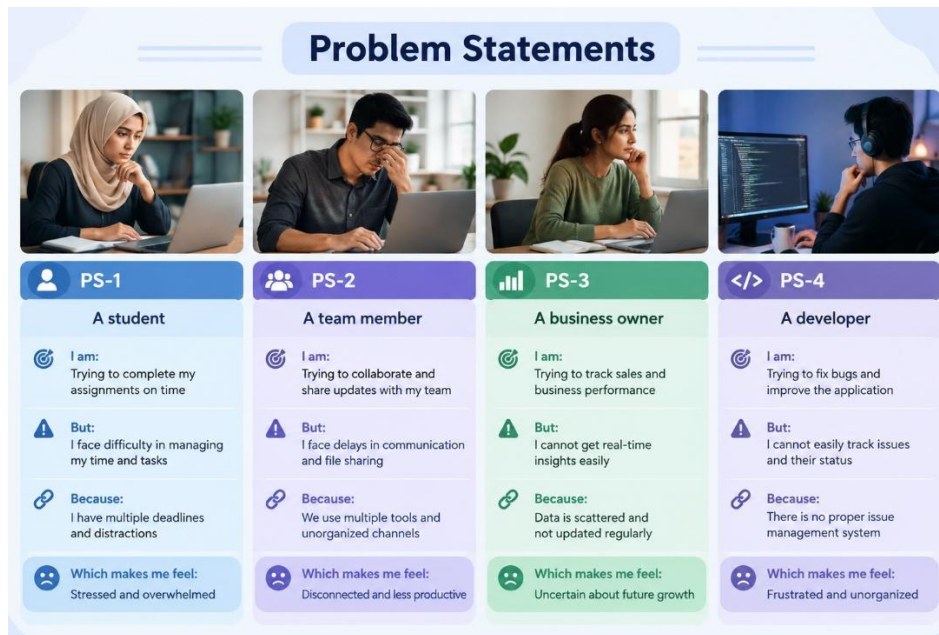
The primary objectives of this project are:

- To establish a centralized platform for managing cricket player performance records.
- To improve the accuracy, consistency, and accessibility of performance data.
- To provide real-time analytical insights through reports and dashboards.
- To support team management, player evaluation, and performance monitoring.
- To automate routine data management processes using Salesforce features.
- To facilitate data-driven decision-making for coaches, analysts, and team administrators.
- To enhance collaboration among team members through a unified performance tracking system.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cricket coach	Evaluate player performance for upcoming matches	I find it difficult to access consolidated performance records	Player statistics are scattered across multiple sources	Uncertain while selecting the best playing squad
PS-2	A team administrator	Manage and maintain player information efficiently	Updating player records consumes significant time	The process relies heavily on manual record keeping	Overburdened and less productive
PS-3	A performance analyst	Identify player strengths, weaknesses, and trends	Comparing performance data across matches is challenging	There is no unified system for performance analysis	Dissatisfied with the quality of insights generated
PS-4	A cricket academy manager	Monitor the progress of players over time	I cannot easily track historical performance data	Records are not stored in a structured manner	Concerned about evaluating player development accurately



2.2 Empathy Map Canvas

Target Users

- Team Managers
- Coaches
- Performance Analysts
- Cricket Academy Administrators

Empathy Analysis

Aspect	Description
Think & Feel	Need reliable player statistics, performance insights, and effective team management tools
Hear	Feedback from coaches, players, selectors, and management regarding team performance
See	Disorganized records, multiple spreadsheets, and scattered performance data
Say & Do	Record match statistics, monitor player progress, and prepare performance reports
Pain	Time-consuming manual data entry, inconsistent records, and difficulty in analyzing player performance
Gain	Centralized data management, automated reporting, real-time insights, and faster decision-making

Problem Statements			
			
 PS-1	 PS-2	 PS-3	 PS-4
A student	A team member	A business owner	A developer
 I am: Trying to complete my assignments on time	 I am: Trying to collaborate and share updates with my team	 I am: Trying to track sales and business performance	 I am: Trying to fix bugs and improve the application
 But: I face difficulty in managing my time and tasks	 But: I face delays in communication and file sharing	 But: I cannot get real-time insights easily	 But: I cannot easily track issues and their status
 Because: I have multiple deadlines and distractions	 Because: We use multiple tools and unorganized channels	 Because: Data is scattered and not updated regularly	 Because: There is no proper issue management system
 Which makes me feel: Stressed and overwhelmed	 Which makes me feel: Disconnected and less productive	 Which makes me feel: Uncertain about future growth	 Which makes me feel: Frustrated and unorganized

2.3 Brainstorming

Brainstormed Ideas

- Cricket Player Performance Tracking System
- Team and Player Statistics Management Platform
- Salesforce-Based Sports Analytics Application
- Performance Dashboard and Reporting System
- Automated Match Statistics Tracking System
- Player Ranking and Comparison Platform
- AI-Assisted Performance Analysis Solution

Selected Idea:

Salesforce-Based Cricket Player Performance Tracking Application (CPPTA)

This keeps the same format as the sample document while making it specific to your project and avoiding direct copying.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login to the CRICKSTATS Application
- Register and Manage Player Information
- Record Match Statistics
- Update Player Performance Records
- View Reports and Dashboards
- Analyze Player and Team Performance
- Compare Player Statistics
- Make Team Selection and Strategic Decisions

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration and Management
FR-2	Match Statistics Entry
FR-3	Performance Tracking and Analysis
FR-4	Report Generation
FR-5	Dashboard Visualization
FR-6	Flow Automation
FR-7	Player Ranking and Comparison

Non-Functional Requirements (NFRs) – CRICKSTATS

NFR-1: Usability

The system should be easy to learn and use by coaches, team managers, analysts, and administrators without requiring extensive technical knowledge.

Requirements

- User-friendly interface
- Easy navigation between objects and records
- Interactive dashboards
- Minimal training requirements

Example

A coach should be able to log in, update player statistics, and view performance reports within a few minutes.

Example: Salesforce Lightning Experience provides intuitive navigation and responsive user interfaces.

NFR-2: Security

The system must ensure that player information and performance records are protected from unauthorized access.

Requirements

- Role-based access control
- Secure user authentication
- Data privacy protection
- Regular system backups

Examples

- Coaches can view player records.
- Team managers can create and edit records.
- Analysts can generate reports.
- Only administrators can manage application settings.

Example: Salesforce Profiles, Roles, and Permission Sets control access to sensitive information.

NFR-3: Reliability

The application should consistently provide accurate information and operate without unexpected failures.

Requirements

- Accurate data storage
- Automatic backup mechanisms
- Error handling and recovery
- High system stability

Example

If a user loses connectivity while entering match statistics, the system should preserve data and synchronize changes when the connection is restored.

Example: Salesforce cloud infrastructure ensures reliable data recovery and availability.

NFR-4: Performance

The system should provide quick responses and process large amounts of player data efficiently.

Requirements

- Fast page loading
- Quick report generation
- Responsive dashboards
- Low processing delays

Example

A report containing performance data for hundreds of players should load within a few seconds.

Example: Salesforce reporting tools and optimized queries improve overall performance.

NFR-5: Availability

The application should be accessible whenever users need to access player information.

Requirements

- 24/7 accessibility
- Minimal downtime
- Cloud-based availability
- Disaster recovery support

Example

Team managers should be able to access player statistics during matches, tournaments, or training sessions at any time.

Example: Salesforce cloud services provide high availability and reliable access.

NFR-6: Scalability

The system should support growth in the number of players, teams, matches, and users.

Requirements

- Support increasing data volumes
- Handle multiple teams and tournaments
- Expand storage as needed
- Maintain performance under heavy workloads

Example

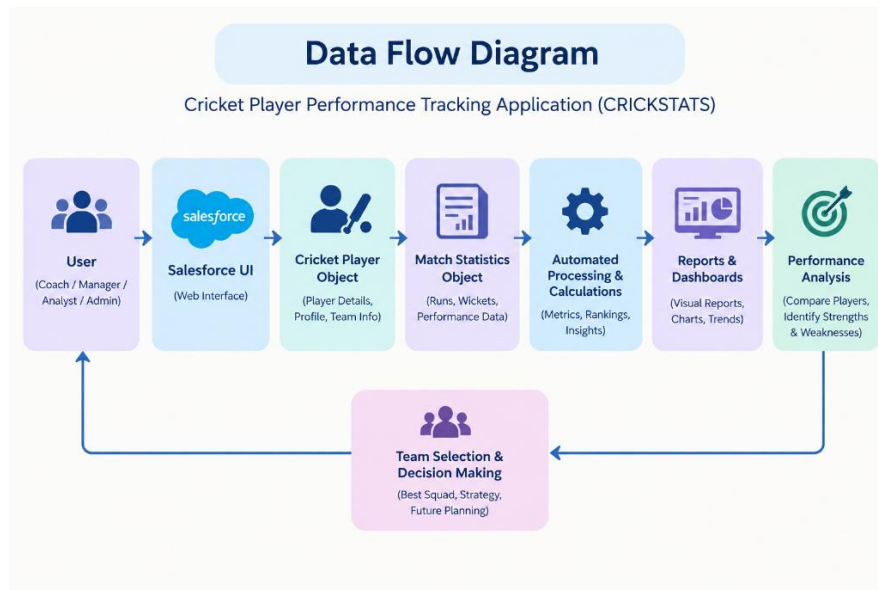
If the application expands from a single team to multiple leagues and tournaments, it should continue operating efficiently.

Example: Salesforce cloud architecture automatically scales resources based on demand.

3.3 Data Flow Diagram

Data Flow

User → Salesforce UI → Cricket Player Object → Match Statistics Object → Reports → Dashboards → Performance Analysis



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning Experience
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Custom Objects & Relationships
Database	Salesforce Cloud Database

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The **Cricket Player Performance Tracking Application (CRICKSTATS)** is designed to address the challenges associated with managing and analyzing cricket player performance data. Traditionally, player statistics, match records, and performance reports were maintained manually using spreadsheets and paper-based records, making the process time-consuming

and prone to errors. By implementing a Salesforce-based solution, the application provides a centralized platform for storing and managing player information efficiently.

One major challenge faced by coaches and team managers is the difficulty of accessing accurate and up-to-date player statistics. CRICKSTATS resolves this issue through structured custom objects and automated data management, ensuring that performance records remain consistent and easily accessible. The application also minimizes the risk of data loss by utilizing Salesforce cloud storage, which provides secure data backup and reliable record management.

Another common problem is the lack of effective performance analysis due to scattered and unorganized data sources. CRICKSTATS addresses this challenge through interactive reports and dashboards that transform raw match statistics into meaningful insights. Coaches, analysts, and administrators can quickly evaluate player performance, compare statistics, and identify strengths and areas for improvement.

Furthermore, the application reduces the time required to generate reports and performance summaries through Salesforce automation features. Real-time access to player statistics enables faster decision-making for team selection, player development, and strategic planning. As a result, CRICKSTATS successfully bridges the gap between traditional manual record-keeping methods and modern data-driven sports management practices.

4.2 Proposed Solution

Description

CRICKSTATS provides:

- Custom Objects (Cricket Player, Match Statistics)
- Master-Detail Relationships
- Roll-Up Summary Fields
- Salesforce Lightning Application
- Automated Salesforce Flows
- Interactive Reports and Dashboards

Key Features

Centralized Database

- Stores all player and match information in a single platform.
- Eliminates duplicate and scattered records.
- Provides quick access to historical performance data.

Live Statistics

- Updates player performance records in real time.
- Tracks runs, wickets, matches played, and other key metrics.
- Supports instant performance analysis.

Secure Access

- Implements role-based access control.
- Protects sensitive player information.
- Ensures secure authentication and data management.

Scalable Design

- Supports increasing numbers of players, teams, and matches.
- Maintains system performance as data grows.
- Easily adaptable for tournaments and multiple leagues.

Easy Reporting

- Generates performance reports automatically.
- Provides visual dashboards and charts.
- Enables quick comparison of player statistics and trends.

4.3 Solution Architecture

Architecture Layers

Presentation Layer

- Salesforce Lightning User Interface
- Lightning App
- Record Pages and Navigation Tabs

Application Layer

- Salesforce Flows
- Business Logic
- Validation Rules
- Automation Processes

Data Layer

- Cricket Player Object
- Match Statistics Object
- Custom Fields and Relationships
- Roll-Up Summary Fields

Analytics Layer

- Reports
- Dashboards
- Performance Charts
- Player Comparison Analytics

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile-based execution
- Incremental feature development
- Sprint-wise implementation
- Continuous testing and feedback

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS application.	1	High	Member1
Sprint-1	Data Modelling	USN-2	As an admin, I want to create Cricket Player and Match Statistics objects so that player data can be stored systematically.	3	High	Member2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for Player and Match Statistics objects so that I can access records easily.	3	High	Member3
Sprint-2	UI Design	USN-4	As a coach, I want a Lightning App so that all	3	High	Member4

			cricket performance features are available in one place.			
Sprint-2	Data Structure	USN-5	As an admin, I want to create custom fields for players and match statistics so that complete performance information can be captured.	5	Medium	Member1
Sprint-2	Page Layout	USN-6	As a user, I want customized page layouts so that relevant information is displayed efficiently.	5	High	Member2
Sprint-3	Validation	USN-7	As a system, I want validation rules to prevent incorrect player and match data entry.	3	High	Member3
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics and performance records automatically.	5	High	Member4
Sprint-4	Analytics	USN-9	As an analyst, I want reports and dashboards so that player performance can be analyzed visually.	5	High	Member1

Sprint-4	Testing	USN-10	As a project team, we want to test the application to ensure accurate functionality and performance.	4	Medium	Member2
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Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 Sep 2026	06 Sep 2026	20	06 Sep 2026
Sprint-2	20	6 Days	08 Sep 2026	13 Sep 2026	20	13 Sep 2026
Sprint-3	20	6 Days	15 Sep 2026	20 Sep 2026	20	20 Sep 2026
Sprint-4	20	6 Days	22 Sep 2026	27 Sep 2026	20	27 Sep 2026

6. PROJECT DEVELOPMENT PHASE

Project Flow

Milestone 1: Creation of Developer Account

Milestone 2: Cricket Player Object Creation

Milestone 3: Match Statistics Object Creation

Milestone 4: Custom Tabs Creation

Milestone 5: Lightning Application Setup

Milestone 6: Fields and Relationships Configuration

Milestone 7: Roll-Up Summary Fields

Milestone 8: Validation Rules

Milestone 9: Salesforce Flows

Milestone 10: Reports and Dashboards

Milestone 11: Testing and Deployment

Implementation

Milestone 1 – Salesforce Developer Account Creation

Activity 1: Creating Developer Account

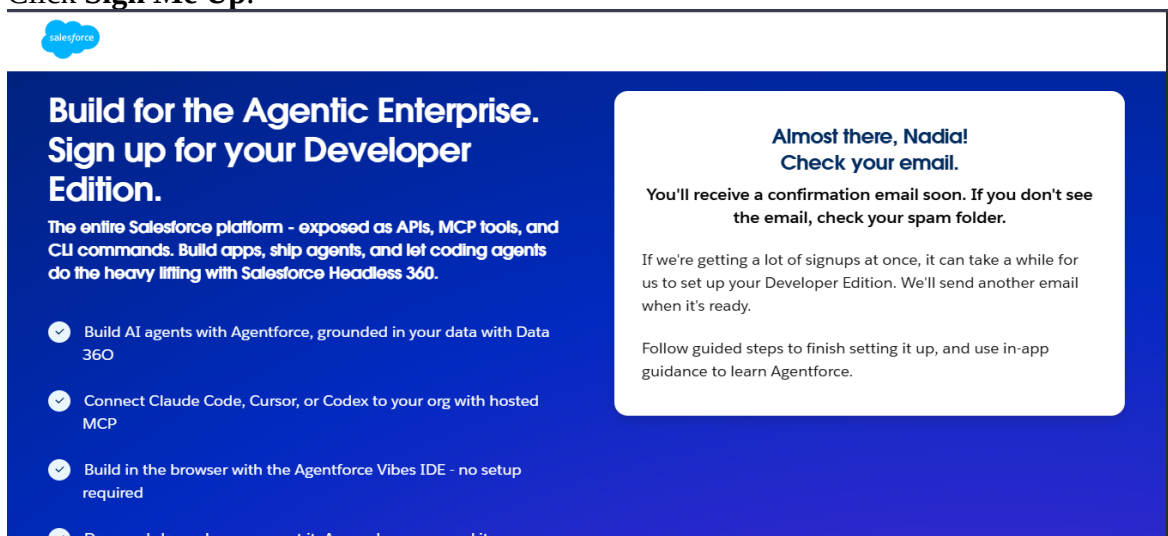
Create a Salesforce Developer Organization for building and testing the CRICKSTATS application.

1. Visit the Salesforce Developer Signup Page:

[Salesforce Developer Signup Page](#)

2. Complete the registration form with the following details:
 - First Name
 - Last Name
 - Email Address
 - Role: Developer
 - Company / College Name
 - Country: India
 - Postal Code
 - Username (must be unique and in email format)

Click **Sign Me Up**.



Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it. Apex when you need it

Almost there, Nadia! Check your email.

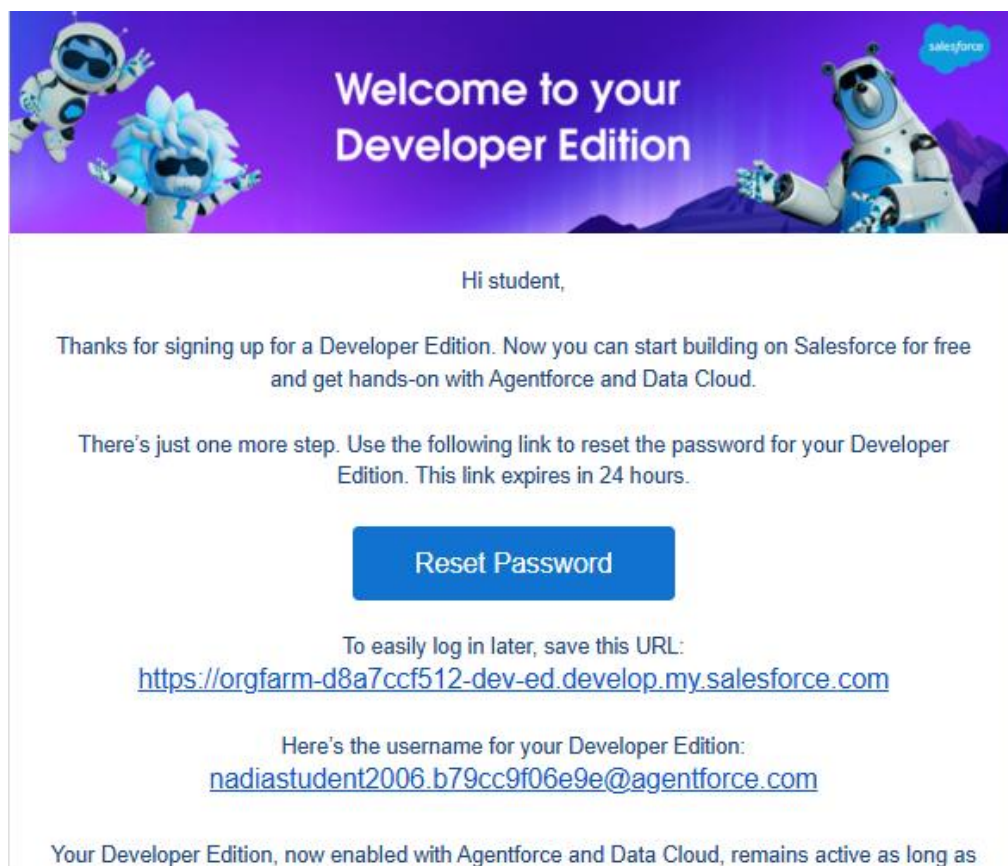
You'll receive a confirmation email soon. If you don't see the email, check your spam folder.

If we're getting a lot of signups at once, it can take a while for us to set up your Developer Edition. We'll send another email when it's ready.

Follow guided steps to finish setting it up, and use in-app guidance to learn Agentforce.

Activity 2: Account Activation

1. Open the verification email received from Salesforce.
2. Click **Verify Account**.
3. Create a password and security question.
4. Click **Change Password**.
5. Login to your Salesforce Developer Org.
6. Access the Salesforce Setup page and begin application development.



Then continue with:

- Milestone 2: Cricket Player Object Creation
- Milestone 3: Match Statistics Object Creation
- Milestone 4: Custom Tabs Creation
- Milestone 5: Lightning App Creation
- Milestone 6: Fields & Relationships
- Milestone 7: Roll-Up Summary Fields
- Milestone 8: Flows
- Milestone 9: Reports & Dashboards

Milestone 2 - Object Creation

Activity 1: Cricket Player Object

1. In your Salesforce org, click the Gear icon and select **Setup**.
2. Click the **Object Manager** tab. If you do not see it, enter **Object Manager** in the Quick Find box.
3. On the Object Manager page, click **Create** → **Custom Object**.
4. From the Setup page → Click **Object Manager** → Click **Create** → Click **Custom Object**.

- **Label:** Cricket Player

Plural Label: Cricket Players

Visualforce tabs allow you to embed Visualforce pages. Lightning component tabs allow you to add Lightning components to the nav bar. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs			New	What Is This?
Action	Label	Tab Style		
Edit Del	Match Performances	 Compass		
Edit Del	Players	 Balls		

Web Tabs			New	What Is This?
No Web Tabs have been defined				

Enter Record Name Label and Format:

Record Name: Player Name

Data Type: Text

- **In Optional Features:** Check the boxes for:
 - Allow Reports
 - Allow Activities
 - Track Field History
- **Search Status:** Check the box for **Allow Search**
- Click **Save**

Activity 2: Match Statistics Object

1. From the Setup page → Click **Object Manager** → Click **Create** → Click **Custom Object**.
- **Label:** Match Statistics
 - **Plural Label:** Match Statistics

Enter Record Name Label and Format:

Record Name: Match Name

Data Type: Text

- **In Optional Features:** Check the boxes for:
 - Allow Reports
 - Allow Activities
 - Track Field History
- **Search Status:** Check the box for **Allow Search**

Click **Save**

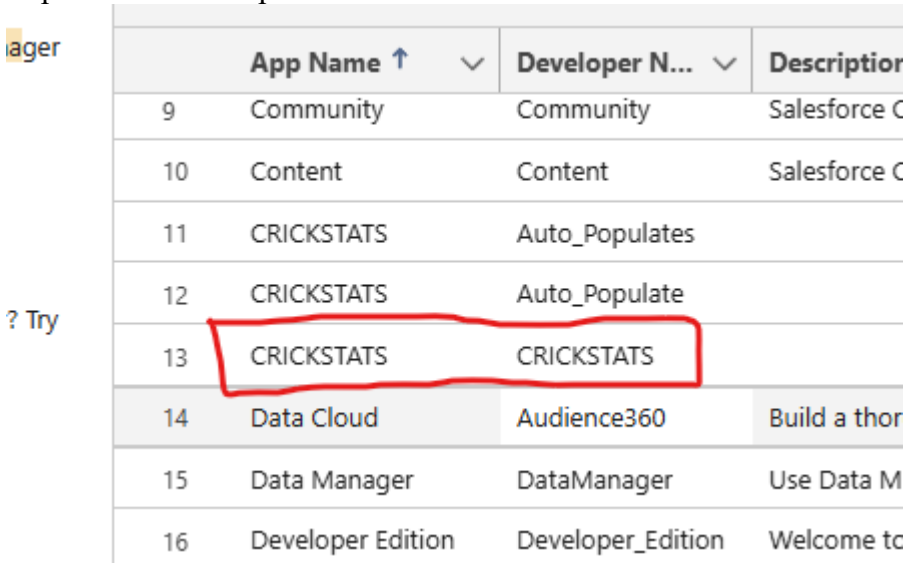
The screenshot shows the Salesforce Setup interface in a web browser. The address bar displays the URL: `orgfarm-d342ec5163-dev-ed.develop.my.salesforce-setup.com/lightning/setup/Objec`. The browser's bookmark bar includes links to Pinterest, ChatGPT, Canva, Gemini, kaaju, WhatsApp, and a folder named 'farhan bhaiya's proj...'. The Salesforce interface features a 'Developer Edition' badge, a search bar labeled 'Search Setup', and a navigation menu with 'Setup', 'Home', and 'Object Manager'. The 'Object Manager' section is active, displaying a header with the 'SETUP' icon, the title 'Object Manager', and the text '53+ Items, Sorted by Label'. Below this is a table with columns 'LABEL', 'API NAME', and 'TYPE'. The table lists several standard objects, including Account, Activity, Activity History, Address, Agent Work, Alternative Payment Method, API Anomaly Event Store, Appointment Category, and Appointment Invitation.

LABEL	API NAME	TYPE
Account	Account	Standard Object
Activity	Activity	Standard Object
Activity History	DevopsActivityLog	Standard Object
Address	Address	Standard Object
Agent Work	AgentWork	Standard Object
Alternative Payment Method	AlternativePaymentMethod	Standard Object
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object
Appointment Category	AppointmentCategory	Standard Object
Appointment Invitation	AppointmentInvitation	Standard Object

Milestone 3 - Tabs

Activity 1: Custom Tabs Creation

1. Go to **Setup** → Type **Tabs** in the Quick Find box → Click **Tabs** → Click **New** under Custom Object Tabs.
2. Select Object (**Cricket Player**) → Select a Tab Style → Click **Next**.
3. On the Add to Profiles page, keep the default settings → Click **Next**.
4. On the Add to Custom Apps page, uncheck **Include Tab** if required → Click **Next**.
5. Ensure **Append tab to users' existing personal customizations** is checked.
6. Click **Save**.
7. Repeat the same steps for **Match Statistics**.

8. 

	App Name ↑	Developer N...	Description
9	Community	Community	Salesforce C
10	Content	Content	Salesforce C
11	CRICKSTATS	Auto_Populates	
12	CRICKSTATS	Auto_Populate	
13	CRICKSTATS	CRICKSTATS	
14	Data Cloud	Audience360	Build a thor
15	Data Manager	DataManager	Use Data M
16	Developer Edition	Developer_Edition	Welcome to

Milestone 4 - The Lightning App

Activity 1: Create a Lightning App

1. Go to **Setup** → Search **App Manager** in Quick Find → Select **App Manager** → Click **New Lightning App**.
2. Fill the App Details and Branding as follows:

App Name: CRICKSTATS

Developer Name: Auto Populates

App Description: Cricket Player Performance Tracking Application

Logo: Optional

Primary Color: Default or custom color

Click **Next**.

App Options

- **Navigation Style:** Standard Navigation
- **Setup Experience:** Setup
- **Supported Form Factors:** Desktop and Phone

Click **Next**.

Utility Items

- **Utility Bar Alignment:** Default

Click **Next**.

Navigation Items

Select the created custom objects and required standard objects:

- Cricket Players
- Match Statistics
- Reports
- Dashboards

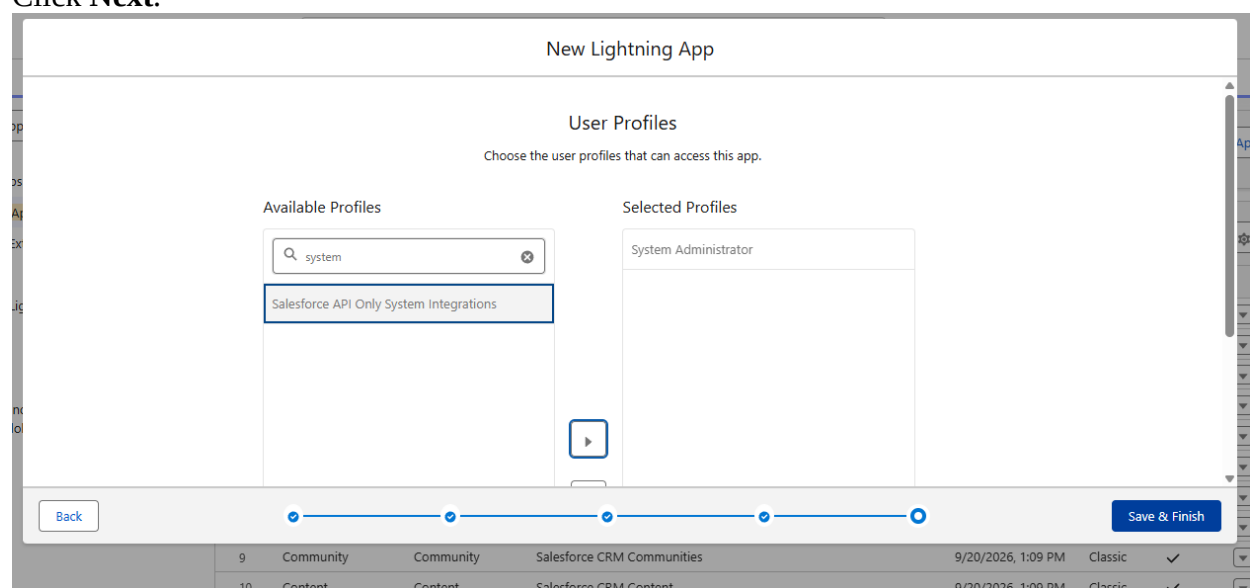
Click **Next**.

User Profiles

Add the following profile:

- System Administrator

Click **Next**.



Review & Finish

1. Review the application settings.
2. Click **Save & Finish**.
3. Launch the **CRICKSTATS** application from the App Launcher.

Expected Outcome



The screenshot shows the Salesforce App Launcher interface. On the left, there are icons for 'App Launcher' and 'Try'. The main area displays a table of applications. The table has four columns: 'App Name', 'Developer Name', and 'Description'. The application 'CRICKSTATS' is highlighted with a red box.

	App Name ↑	Developer N...	Description
9	Community	Community	Salesforce C
10	Content	Content	Salesforce C
11	CRICKSTATS	Auto_Populates	
12	CRICKSTATS	Auto_Populate	
13	CRICKSTATS	CRICKSTATS	
14	Data Cloud	Audience360	Build a thor
15	Data Manager	DataManager	Use Data M
16	Developer Edition	Developer_Edition	Welcome tc

The CRICKSTATS Lightning Application should now contain:

- Cricket Players Tab
- Match Statistics Tab
- Reports Tab
- Dashboards Tab

Milestone 5: Fields & Relationships

The CRICKSTATS application requires a structured data model to efficiently store, manage, and analyze cricket player information and match statistics. To achieve this, custom fields are created within the Player and Match Performance objects. These fields capture essential details such as player profiles, batting and bowling information, match records, and performance statistics. Relationships between the objects are also established to maintain data integrity and enable advanced reporting, roll-up summaries, and analytics. The following activities describe the creation of custom fields and relationships used in the application.

Activity 1: Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.

3. From the object manager page, In the Quick Find box, Search for the custom object you just created: **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.
6. Next, choose a data type as Text. Choosing a data type helps you format the field input.

image

7. Click on next.

8. Data Type: **Text**

Field Label: Player Name

Field Name: Auto Populates

Length: 50

Click on Next → Next → Save and new.

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Player Number
Data Type	Text
Field Name	Player_Number
Description	

Select the page layouts that should include this field. The field will be added to the layout.

9. Repeat the Same steps for remaining fields.

- Data Type: **Text**

Field Label: Player Number

Field Name: Auto Populates

Length: 3

Click on Next → Next → Save and new.

- Data Type: **Rich Text Area**

Field Label: Profile Picture
Field Name: Auto Populates
Click on Next → Next → Save and new.

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Profile Picture
Data Type	Rich Text Area
Field Name	Profile_Picture
Description	

Select the page layouts that should include this field. The field will be available on the selected layouts.

- Data Type: **Date**

Field Label: Date of Birth
Field Name: Auto Populates
Click on Next → Next → Save and new.

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Date of Birth
Data Type	Date
Field Name	Date_of_Birth
Description	

Select the page layouts that should include this field. The field will be available on the selected layouts.

- Data Type: **Picklist**

Field Label: Role

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Batsman

Bowler

Batsman & Wicket Keeper

All Rounder

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Role
Data Type	Picklist
Field Name	Role
Description	

Select the page layouts that should include this field. The field will be added to the selected layouts.

- Data Type: **Picklist**

Field Label: Batting Style

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Right-Handed

Left-Handed

Click on Next → Next → Save and new.

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Batting Style
Data Type	Picklist
Field Name	Batting_Style
Description	

- Data Type: **Picklist**

Field Label: Bowling Style

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Right-Arm Fast

Left-Arm Fast

Right-Arm Medium

Left-Arm Medium

Right-Arm Spin

Left-Arm Spin

Click on Next → Next → Save and new.

The screenshot shows the 'New Custom Field' page in Salesforce. At the top, it says 'Player' and 'New Custom Field'. Below this is a blue header bar that says 'Step 4. Add to page layouts'. Underneath, there is a table with the following details:

Field Label	Bowling Style
Data Type	Picklist
Field Name	Bowling_Style
Description	

Below the table, there is a text prompt: 'Select the page layouts that should include this field. The field will be added as the last'.

- Data Type: **Picklist**

Field Label: Nationality

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Australia

Bangladesh

England

India

New Zealand

Pakistan

South Africa

Sri Lanka

West Indies

Afghanistan

Zimbabwe
Ireland
Scotland
Netherlands
United Arab Emirates
Nepal
United States of America

Click on Next → Next → Save and new.

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Nationality
Data Type	Picklist
Field Name	Nationality
Description	

Select the page layouts that should include this field. The field will be

- Data Type: **Picklist**

Field Label: Team

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Chennai Super Kings (CSK)
Delhi Capitals (DC)
Kolkata Knight Riders (KKR)
Mumbai Indians (MI)
Punjab Kings (PBKS)
Rajasthan Royals (RR)
Royal Challengers Bangalore (RCB)
Sunrisers Hyderabad (SRH)
Gujarat Titans (GT)

Lucknow Super Giants (LSG)

Player

New Custom Field

Step 4. Add to page layouts

Field Label	Team
Data Type	Picklist
Field Name	Team
Description	

- Data Type: **Formula**

Field Label: AVG Score

Field Name: Auto Populates

Formula Return: **Number**

Player

New Custom Field

Step 5. Add to page layouts

Field Label	AVG Score
Data Type	Formula
Field Name	AVG_Score
Description	

Select the page layouts that should include this field. The field will be added as the last field in the first 2-col.

Formula:

IF

Matches_Played__c > 0,
Total_Runs__c / Matches_Played__c,
0

Player
New Custom Field

Step 3. Enter formula

Enter your formula and click Check Syntax to check for errors. Click the Advanced Formula subtab to use additional fields, operators, and functions.

Example: `Fahrenheit = 1.8 * Celsius_c + 32` [More Examples...](#)

Simple Formula | **Advanced Formula**

AVG Score (Number) =
 IF(Matches_Played__c > 0,
 Total_Runs__c / Matches_Played__c ,
 0)

Functions
 -- All Function Categories --
 ABS
 ACOS
 ADDMONTHS
 AND
 ASCII
 ASIN

No syntax errors in merge fields or functions. (Compiled size: 130 characters)

Click on Next → Next → Save & New.

- Data Type: **Lookup Relationship**

Related To: Match Performance

Field Label: Match Performance

Field Name: Auto Populates

Click on Next → Next → Save.

Activity 2: Create Custom fields for Match Performance object

The Match Performance object is used to record detailed statistics of a player for each match. These fields help track batting, bowling, and fielding performance and serve as the foundation for reports, dashboards, and roll-up summaries.

1. Data Type: **Text**

Field Label: Match Performance Name

Field Name: Auto Populates

Length: 80

Click on Next → Next → Save and new.

2. Data Type: **Checkbox**

Field Label: Player of the Match

Field Name: Auto Populates

Click on Next → Next → Save and new.

Match Performance New Custom Field

Step 4. Add to page layouts

Field Label	Player of the Match
Data Type	Checkbox
Field Name	Player_of_the_Match
Description	

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section.

3. Data Type: **Picklist**

Field Label: Match Result

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

WON

LOST

DRAW

Click on Next → Next → Save and new.

4. Data Type: **Number**

Field Label: Runs Scored

Field Name: Auto Populates

Click on Next → Next → Save and new.

5. Data Type: **Number**

Field Label: Balls Faced

Field Name: Auto Populates

Click on Next → Next → Save and new.

6. Data Type: **Number**

Field Label: Fours

Field Name: Auto Populates

Click on Next → Next → Save and new.

7. Data Type: **Number**

Field Label: Sixes

Field Name: Auto Populates

Click on Next → Next → Save and new.

8. Data Type: **Number**

Field Label: Overs Bowled

Field Name: Auto Populates

Click on Next → Next → Save and new.

9. Data Type: **Number**

Field Label: Wickets Taken

Field Name: Auto Populates

Click on Next → Next → Save and new.

10. Data Type: **Number**

Field Label: Runs Conceded

Field Name: Auto Populates

Click on Next → Next → Save and new.

11. Data Type: **Number**

Field Label: Maiden Overs

Field Name: Auto Populates

Match Performance

New Custom Field

Step 4. Add to page layouts

Field Label	Maiden Overs
Data Type	Number
Field Name	Maiden_Overs
Description	

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section.

Click on Next → Next → Save and new.

12. Data Type: **Number**

Field Label: Stumpings

Field Name: Auto Populates

Click on Next → Next → Save and new.

Match Performance

New Custom Field

Step 4. Add to page layouts

Field Label	Stumpings
Data Type	Number
Field Name	Stumpings
Description	

Select the page layouts that should include this field. The field will be added as the last field

13. Data Type: **Number**

Field Label: Catches

Field Name: Auto Populates

Click on Next → Next → Save and new.

Match Performance

New Custom Field

Step 4. Add to page layouts

Field Label	Catches
Data Type	Number
Field Name	Catches
Description	

Select the page layouts that should include this field. The field will be added as the last fi

14. Data Type: **Master-Detail Relationship**

Related To: Player
Field Label: Player
Field Name: Auto Populates

Click on Next → Next → Save.

Match Performance New Relationship

Step 5. Add reference field to Page Layouts

Field Label	Player
Data Type	Master-Detail
Field Name	Player
Description	

These are the page layouts that will include this field. Because this is a Master-Detail relationship, the field is required.

Add Field	Page Layout Name
☒	Match Performance Layout

In this milestone, we focus on creating **Roll-Up Summary fields** on the **Player object** to automatically calculate and aggregate match statistics from the related **Match Performance object**. This ensures that each player's overall career statistics are updated dynamically based on their match records.

Activity 1: Create Roll-Up Summaries for Player Object

Note: Roll-Up Summaries can only be created on the parent object in a Master-Detail Relationship.

Steps:

1. In your Salesforce org, click the **gear icon** → select **Setup**.
2. Navigate to **Object Manager**.
3. Search for the custom object **Player**.
4. From the sidebar, click **Fields & Relationships**.
5. Click **New** → choose **Roll-Up Summary** → click **Next**.
6. Configure the following fields one by one:
 - **Matches Played**
 - Summarized Object: Match Performances
 - Roll-Up Type: COUNT
 - Filter Criteria: All records
 - **Number of 50's**
 - Summarized Object: Match Performances

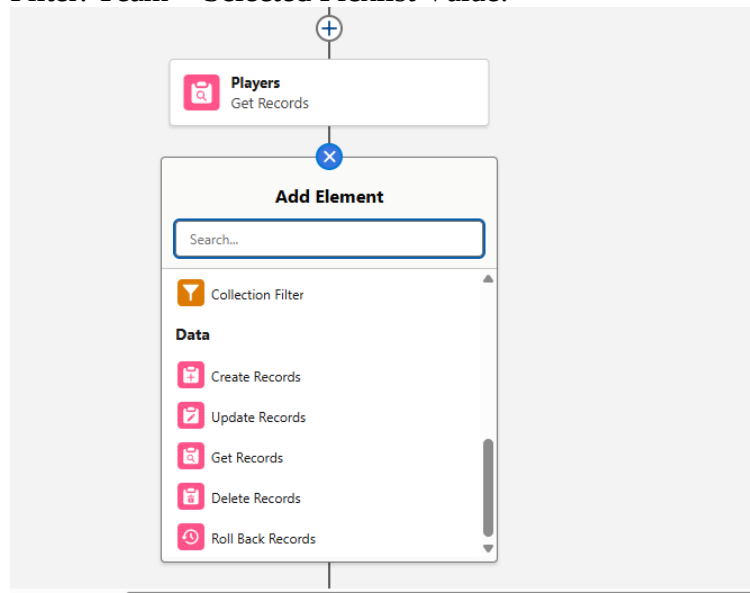
- Roll-Up Type: COUNT
 - Filter Criteria: Records where Runs Scored ≥ 50 and < 100
 - **Number of 100's**
 - Summarized Object: Match Performances
 - Roll-Up Type: COUNT
 - Filter Criteria: Records where Runs Scored ≥ 100
 - **Total 4's**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Fours
 - **Total 6's**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Sixes
 - **Total Runs**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Runs Scored
 - **Total Overs**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Overs Bowled
 - **Total Wickets**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Wickets Taken
 - **Total Stumpings**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Stumpings
 - **Total Catches**
 - Summarized Object: Match Performances
 - Roll-Up Type: SUM
 - Field to Aggregate: Catches
7. After each configuration, click **Next** → **Next** → **Save and New** until all fields are created.

Milestone 7: Flows

Salesforce Flow is used to automate processes and provide an interactive user experience within the **CRICKSTATS** application. The Screen Flow developed in this project allows users to select a team and instantly view corresponding player information, improving accessibility and decision-making.

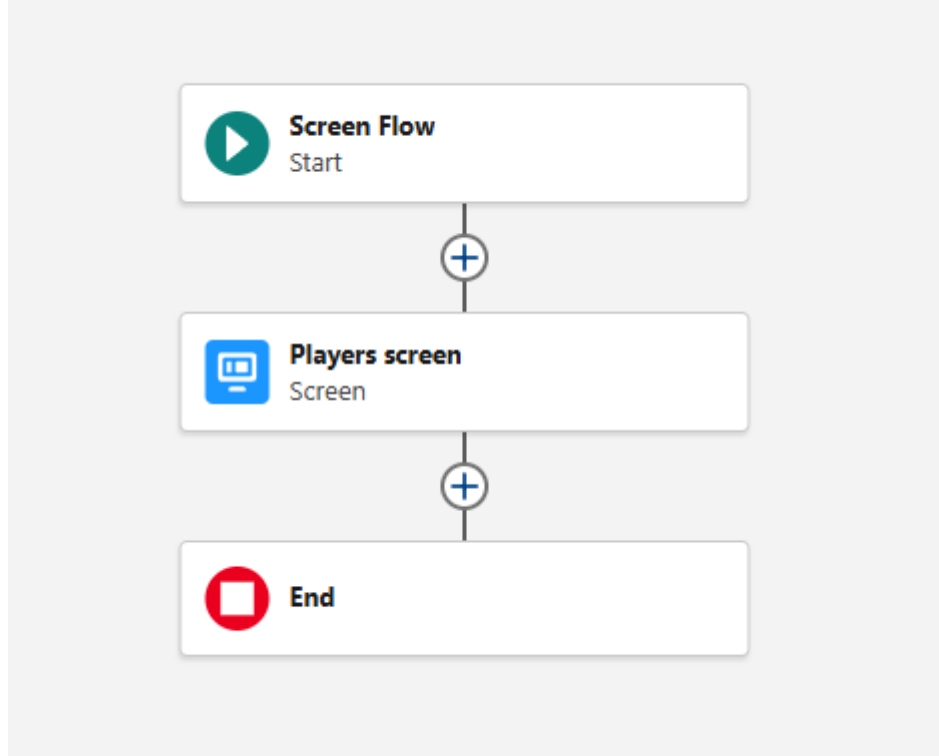
Activity 1: Screen Flow

1. Go to **Setup**, type **Flow** in the Quick Find box, and select **New Flow**.
2. Select **Screen Flow** and click **Create**.
3. Configure the **Team Selection Screen**:
 - Add a **Picklist component** for Team selection.
 - API Name: Team
 - Source: Player Object → Team field.
4. Retrieve player records dynamically using the selected team:
 - Add a **Get Records** element.
 - Object: Player
 - Filter: Team = Selected Picklist Value.



5. Display player information using a **Data Table component**:

- Columns: Profile Picture, Player Name, Total Runs, Average Score, Batting Style, Bowling Style, Matches Played, Number of 50's, Number of 100's.



5. Click (+) Icon on the flow builder and Select **Screen Component**.

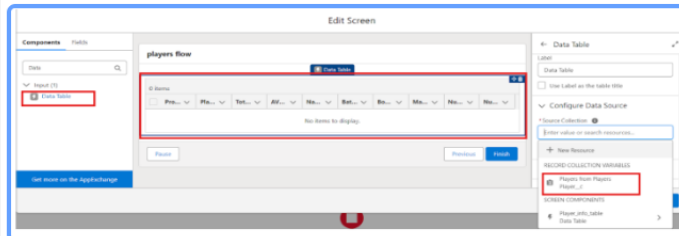
Label : Player_info

API Name : Player_info

Component Pane on the left side select Component Type :

API Name : Player_info_table Label : Data Table

Data Source : Source Collection : Players from players (Player__c)

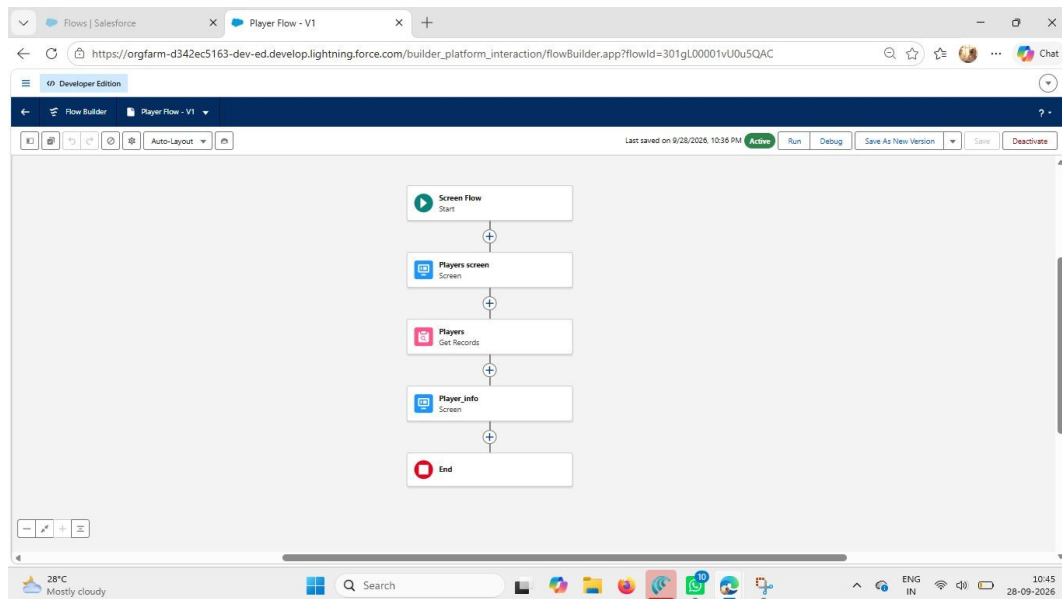


In Configure Columns : Add all the required Columns

Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling Style |
Matches Played | Number of 50's | Number of 100's

Click Done.

-
-



6.

7. Save the flow with:

- **Flow Label:** Player Flow
- **Flow API Name:** Player_Flow

Activity 2: Flow on Home Page

To improve user accessibility, the **Player Flow** is added to the Home Page of the CRICKSTATS application.

Steps:

1. Add **Home Page** to the navigation bar.
2. Open the Home Page for editing.
3. Add the **Flow component**.
4. Select **Player Flow**.
5. Activate the page.

8. Advantages & Disadvantages

Advantages

- **Centralized System:** CRICKSTATS consolidates all cricket data (players, matches, statistics) into a single Salesforce platform, eliminating fragmented spreadsheets or manual records.
- **Real-Time Analytics:** Automated roll-up summaries and dashboards provide instant updates on player and team performance, enabling coaches and analysts to make timely decisions.
- **Secure Platform:** Built on Salesforce Cloud, the system ensures enterprise-grade security with role-based access, encrypted data storage, and compliance with privacy standards.

- **Scalable Design:** The architecture supports growth in both data volume and user base, making it adaptable for larger leagues or multi-team environments.
- **Reduced Manual Work:** Automated flows, summaries, and dashboards minimize repetitive tasks such as calculating averages or compiling reports, saving significant time and effort.

Disadvantages

- **Requires Salesforce License:** Access to CRICKSTATS depends on Salesforce licensing, which may involve additional costs for organizations.
- **Internet Dependency:** As a cloud-based solution, it requires a stable internet connection; performance may be affected in low-connectivity regions.
- **Limited Offline Access:** Users cannot access dashboards or flows without logging into Salesforce, restricting usage in offline scenarios.

9. Conclusion

CRICKSTATS successfully implements a **cloud-based cricket performance tracking system** using Salesforce. By automating data management, the system improves reporting accuracy, reduces manual effort, and supports informed decision-making for coaches, analysts, and fans.

The project demonstrates:

- Practical application of **Software Development Life Cycle (SDLC)** principles.
- Use of **Agile methodology** for iterative development and testing.
- Effective utilization of **CRM-based development** to deliver a sports analytics solution.

Overall, CRICKSTATS bridges the gap between raw cricket data and actionable insights, showcasing how Salesforce can be leveraged beyond traditional business use cases.

10. Future Scope

- **Mobile Application Integration:** Extend CRICKSTATS to mobile devices for on-the-go access to player stats and dashboards.
- **AI-Based Performance Prediction:** Incorporate machine learning models to forecast player performance trends, injury risks, or match outcomes.
- **Video Analysis Integration:** Link match footage with performance data for deeper insights into player techniques and strategies.
- **Player Ranking System:** Develop automated ranking algorithms based on aggregated statistics, enabling comparisons across leagues.
- **Public Statistics Portal:** Create a public-facing portal for fans to explore player and team stats without requiring Salesforce login.
- **Multi-League Support:** Expand the system to handle multiple leagues simultaneously, with separate dashboards and reporting modules.

11. Appendix

- **Salesforce Objects & Fields:**
 - Player Object → Player Name, Role, Batting Style, Bowling Style, Team, Nationality.
 - Match Performance Object → Runs, Wickets, Overs, Catches, Stumpings, Match Date.
 - Roll-Up Summaries → Matches Played, Total Runs, Total Wickets, Number of 50's, Number of 100's, etc.
- **Manual Match Statistics:**
 - Sample records entered for testing flows and dashboards.
 - Includes batting and bowling data for multiple players across matches.
- **Flow Configurations:**
 - Player Flow → Team selection screen, Get Records element, Data Table display.
 - Home Page Integration → Flow component added to CRICKSTATS Home Page for easy access.

Assign the page to the **CRICKSTATS** application.

